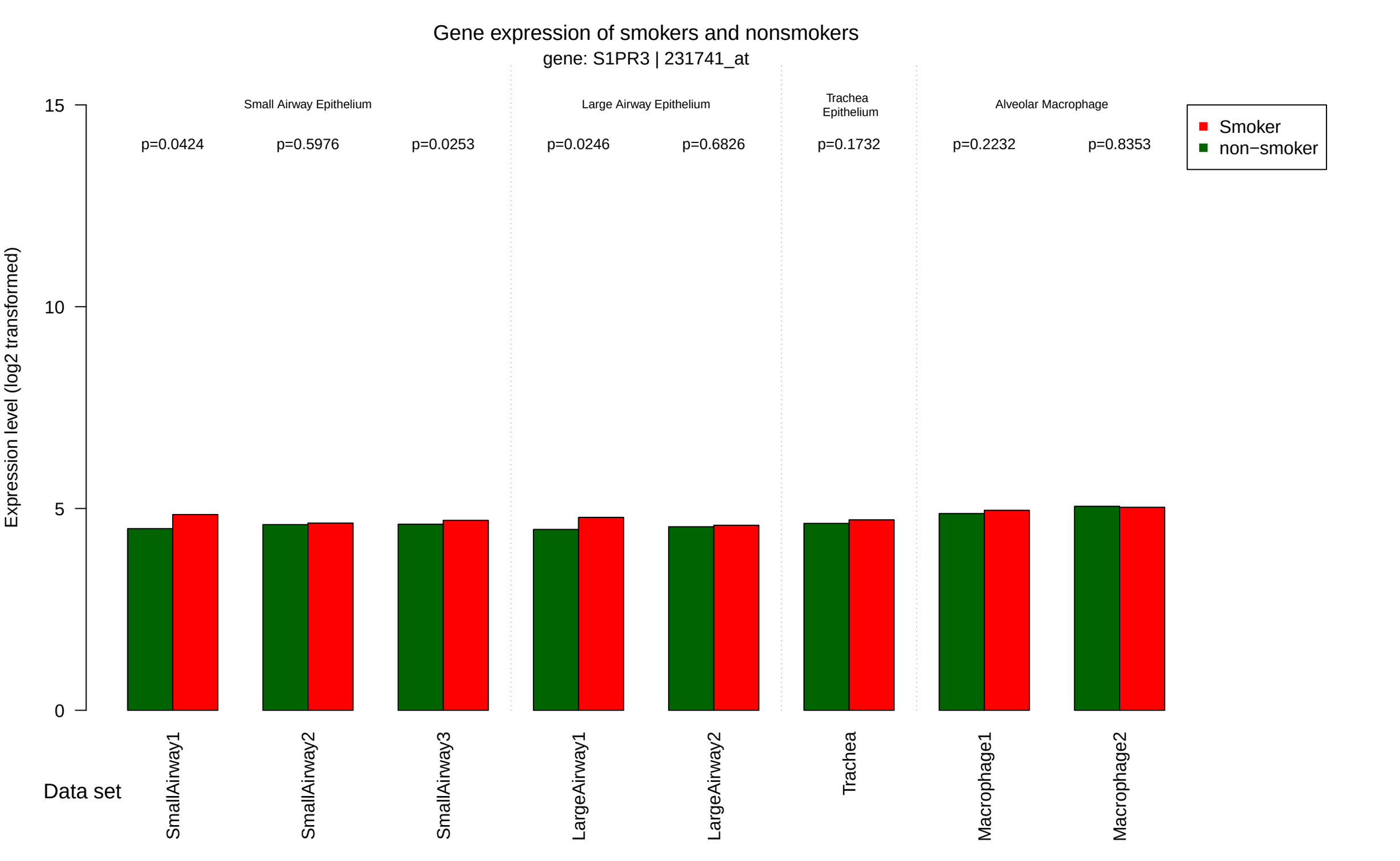
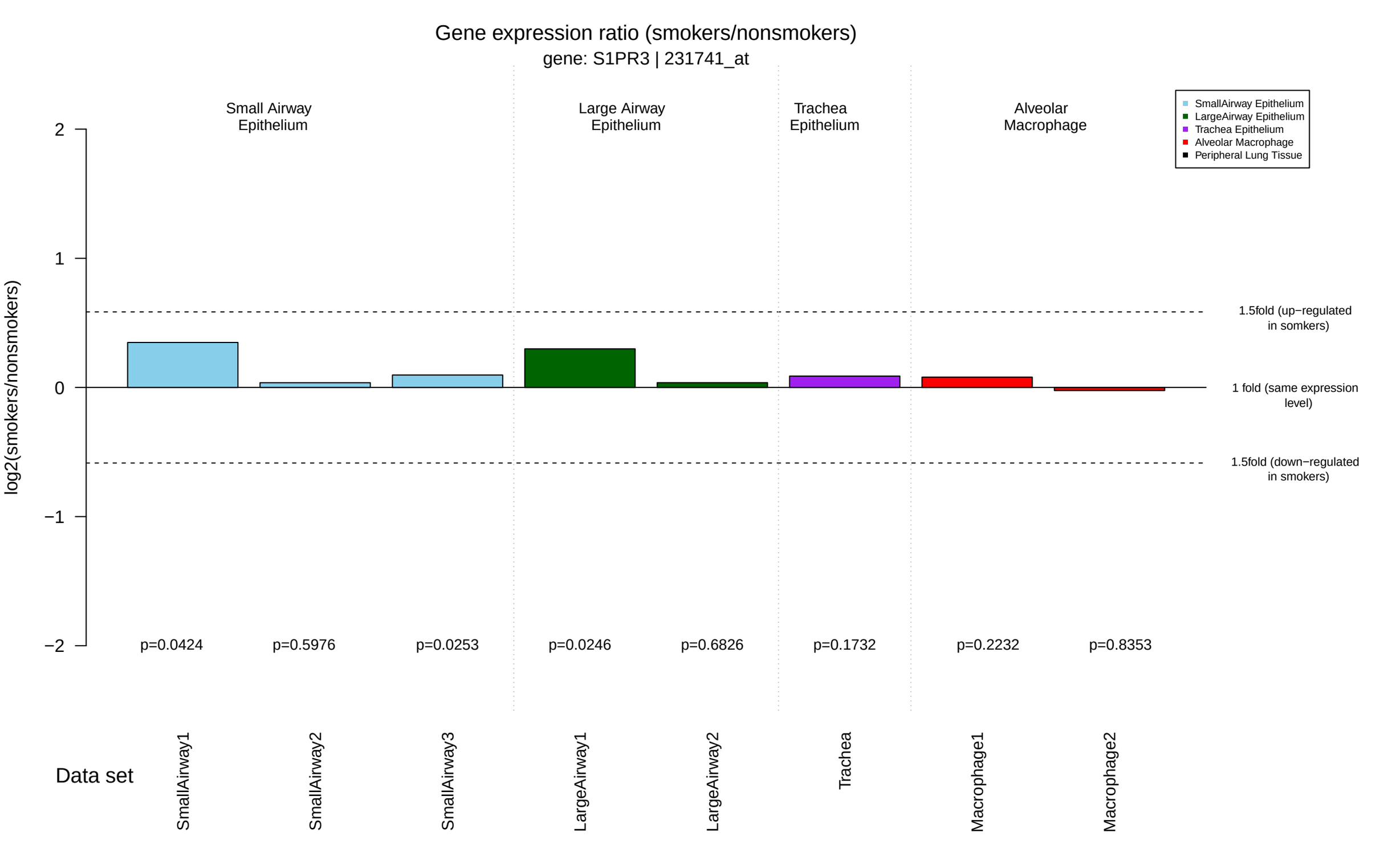
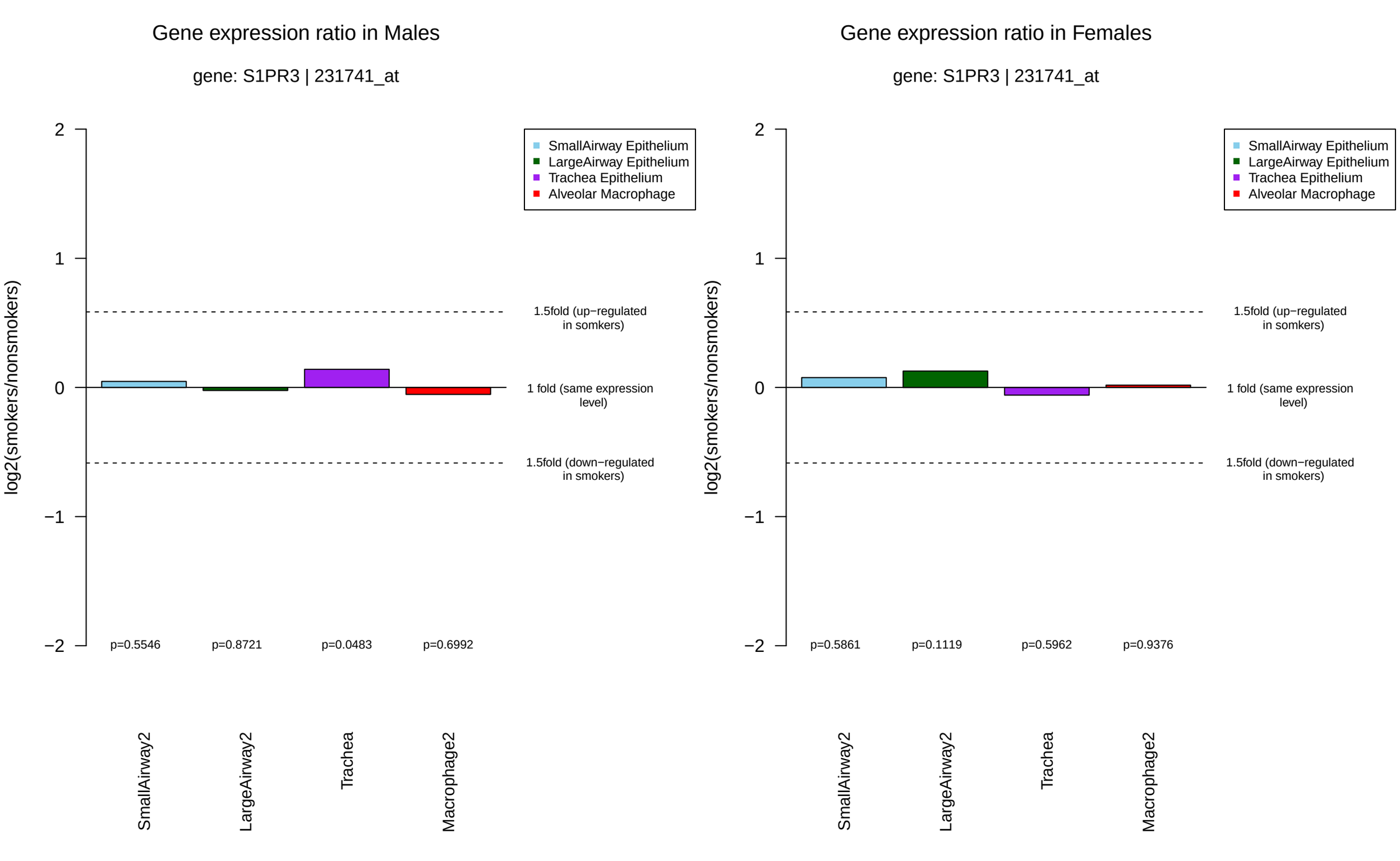
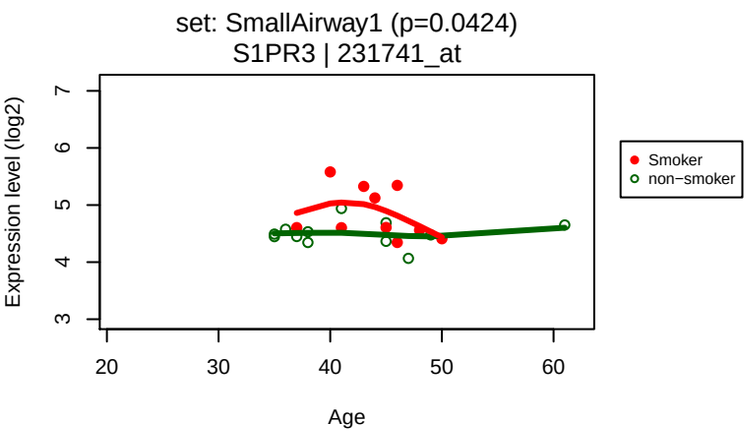
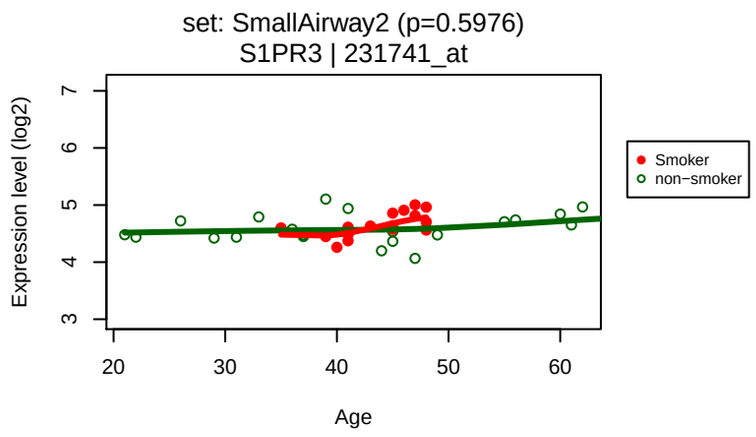
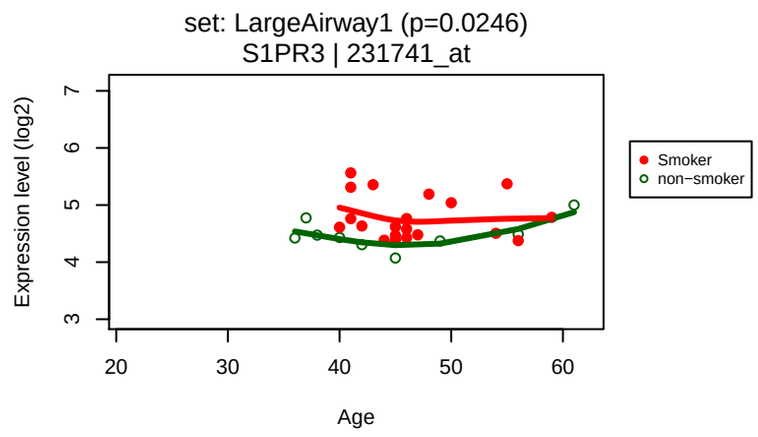

Gene expression of smokers and nonsmokers
gene: S1PR3 | 231741_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0424
p=0.5976
p=0.0253
p=0.0246
p=0.6826
p=0.1732
p=0.2232
p=0.8353
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

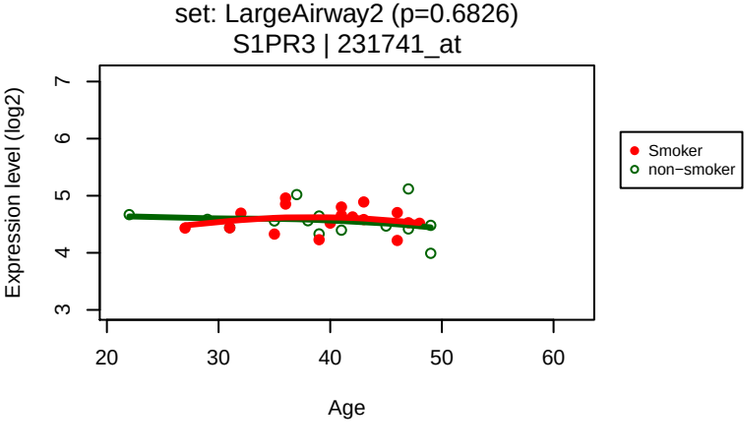

Gene expression ratio (smokers/nonsmokers)
gene: S1PR3 | 231741_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0424
p=0.5976
p=0.0253
p=0.0246
p=0.6826
p=0.1732
p=0.2232
p=0.8353
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

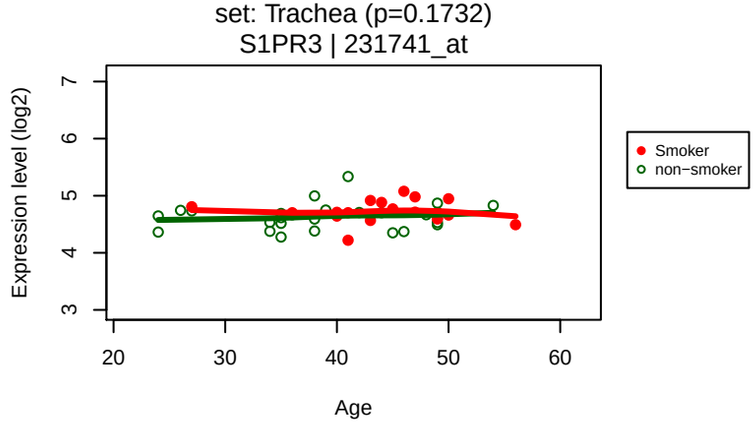

Gene expression ratio in Males
gene: S1PR3 | 231741_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5546
p=0.8721
p=0.0483
p=0.6992
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: S1PR3 | 231741_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5861
p=0.1119
p=0.5962
p=0.9376
SmallAirway2
LargeAirway2
Trachea
Macrophage2

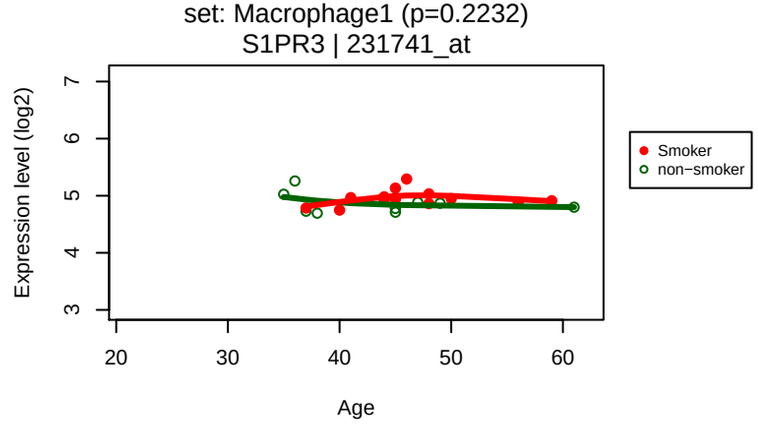

set: SmallAirway1 (p=0.0424)
S1PR3 | 231741_at
Expression level (log2)
Age
Smoker
non−smoker

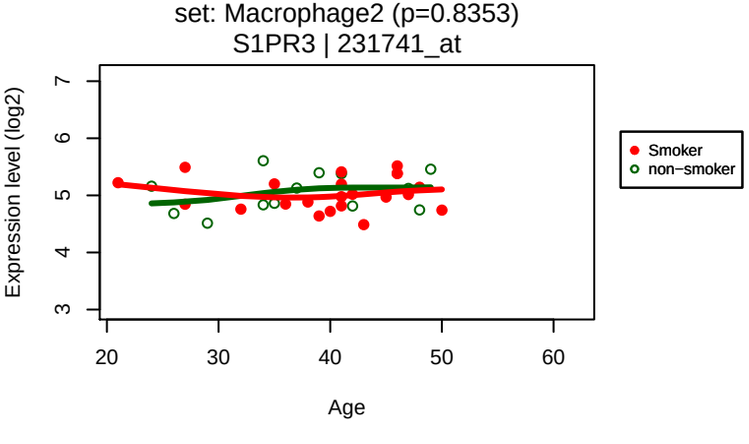

set: SmallAirway2 (p=0.5976)
S1PR3 | 231741_at
Expression level (log2)
Age
Smoker
non−smoker

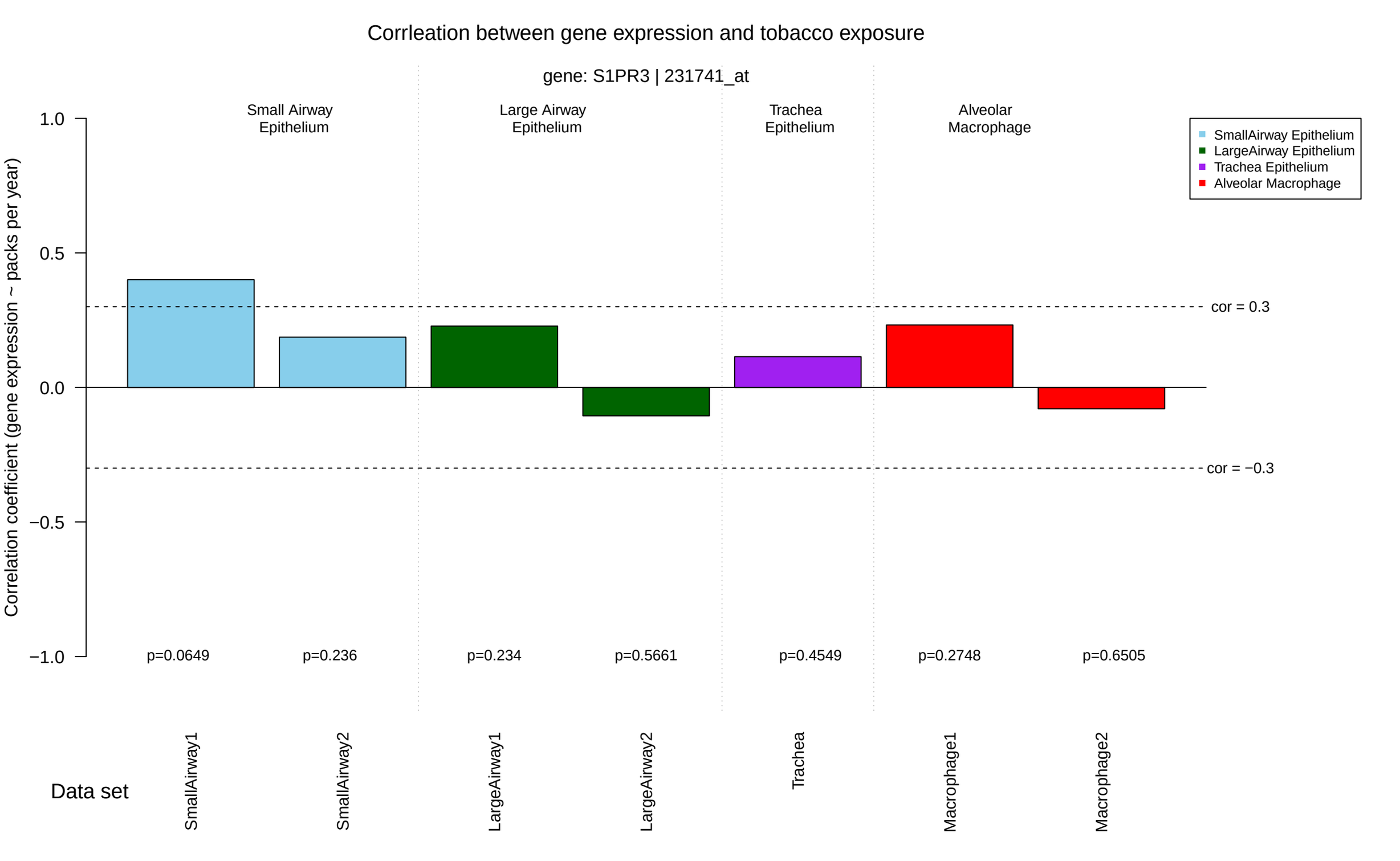

set: LargeAirway1 (p=0.0246)
S1PR3 | 231741_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6826)
S1PR3 | 231741_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1732)
S1PR3 | 231741_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2232)
S1PR3 | 231741_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8353)
S1PR3 | 231741_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: S1PR3 | 231741_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0649
p=0.236
p=0.234
p=0.5661
p=0.4549
p=0.2748
p=0.6505
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2